

# Linux Operating System

## Software

Software is a set of instructions or a program to perform a specific task  
Eg: Calculator, MS Office etc

### Types of Software

1. System Software
2. Application Software

#### \* System Software

It is used to manage the compute hardware

Eg: Operating System, BIOS (Basic Input Output System)

#### \* Application Software

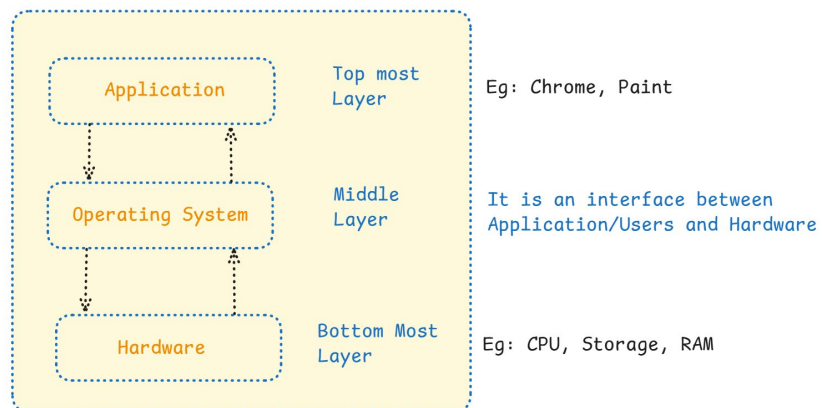
It is a kind of software that perform specific function given by the user  
is called as Application Software

Eg: VLC Media, Calculator, Paint etc

## Architecture of Operating System

There are 3 different layers in Architecture of Operating System

1. Application
2. Operating System
3. Hardware



#### \* Operating System

It is a type of System Software which acts as an interface between application and Hardware

## Functionalities of the Operating System

### Responsibility/Capability of an Operating System

1. Process Management  
Decides the order in which the tasks are executed
2. Memory Management (RAM)  
Allocates and De-allocates the memory needed by the process
3. Device Management  
Enable the communication between the operating system and the hardware devices
4. File Management  
Manage the file creation, Deletion and Organization
5. User Interface  
It is the way that Operating System provides you to communicate with the System

# Introduction to Linux

## Linux Operating System

Linux is a "Kernel Based Free and Open Source Operating System" that will manage the hardware and Software.

## History of Linux

1969 : Unix was developed in the year 1969, It was not free to use and too many versions

1983 : Richard Stallman started a project called as GNU (GNU not UNIX)

1991 : Linus Torvalds developed "Freax", which was later renamed as "LINUX Kernel"

1992 : First version of Linux was released

## Flavours of Linux

### Distributions of Linux

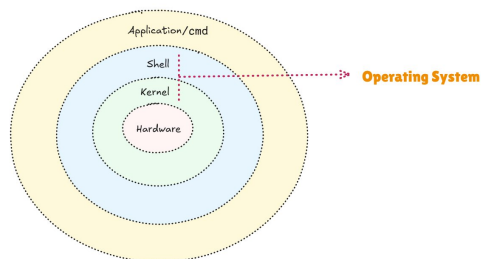
1. Amazon Linux
2. Ubuntu
3. Kali Linux
4. Fedora
5. Debian
6. Cent OS
7. Red Hat
8. Puppy Linux
9. Rocky Linux
10. Parrot OS

Around 500+ distributions of Linux are present in the market

## Features of Linux Operating System

1. Open Source  
Linux is free and Open Source software, Meaning that Linux is available to view, modify and distribute
2. Multitasking  
Linux can run multiple application without affecting the performance
3. Multiuser  
Multiuser can work at the same time without interacting with each other
4. Portability  
Linux can run on various hardware platforms from desktop to laptops, and from servers to supercomputers
5. Stability  
Rarely crashes and run few years without reboot
6. Security  
Linux has strong security features including file permission and user roles

## Architecture of Linux Operating System

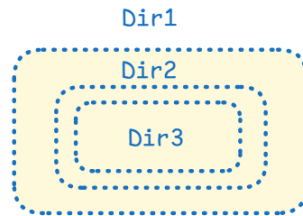


1. Application/CLI/Users  
It is where endusers can use the application
2. Shell  
--> Shell is a command line interface (CLI) where you are going to execute the commands  
--> Shell is responsible for communication between application/user and Kernel
3. Kernel  
--> It is a "Core part" of Linux Operating System  
--> It will establish a communication between shell and hardware  
--> It is program that is stored inside an operating system
4. Hardware  
Hardware is the physical component of the system

| Linux Operating System                                                               | Windows Operating System                                                |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 1. Linux is a free and open source                                                   | --> Windows is paid and closed source                                   |
| 2. It is not user friendly                                                           | --> It is user friendly                                                 |
| 3. It is stable and no need to reboot                                                | --> Windows requires reboot while updating                              |
| 4. It uses more CLI (Command Line Interface) and Less GUI (Graphical User Interface) | --> It uses more GUI and less CLI                                       |
| 5. It is more secured                                                                | --> It is less secured compared to Linux                                |
| 6. It supports multi user and multi tasking                                          | --> It supports less multiuser and less multi tasking compared to Linux |
| 7. It was developed by Linus Torbvalds                                               | --> It is developed by Microsoft                                        |

# Basic Linux Commands

1. `ls -->` (To list all the files and directories)
2. `mkdir <directory_name> -->` (To create a directory)
3. `mkdir Dir1 Dir2 -->` (To create multiple directories at a time)
4. `mkdir -p Dir1/Dir2/Dir3 -->` (This will create a directory inside another directory)



5. `ls -l -->` (It will give long listing format)
6. `clear -->` (It will clear the screen)
7. `touch -->` (It is used to create a file)
8. `pwd -->` (To check the directory where you are working)
9. `cd -->` (It will come to home directory)
10. `rm <file_name> -->` (It is used to delete a file)
11. `rmdir <directory_name> -->` (It will delete an empty directory)
12. `rm -r <directory_name> -->` (It will delete a directory that has items)
13. `ls <directory_name> -->` (To list all the files without getting inside the directory)
14. `vi/vim <file_name> -->` (It is used to add content inside a file)  
Note: 1. `:wq` (write and quit)  
2. `:q!` (quit without making any changes)  
3. `:q` (If not done any changes in the file)
15. `nano <file_name> -->` (It is used to add content inside a file)
16. `cat <file_name> -->` (To read the content of a file)
17. `cat < <file_name> -->` (To read the content of a file)
18. `cat > <file_name> -->` (It is used to override the content)
19. `cat >> <file_name> -->` (It will add new data)
20. `cd .. -->` (It will move one step backward)
21. `cd ../../ -->` (It will move two step backward)
22. `echo "message" -->` (To print any message)
23. `history -->` (To check the history of commands in the terminal)
24. `cal -->` (It will list the calendar)
25. `date -->` (It will show the date)
26. `ls -a -->` (It will show the hidden files)

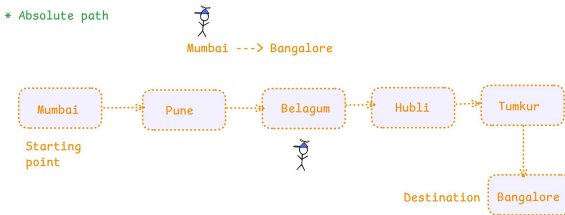
# PATH

It is an exact location of where your files and directories are present

There are two different types of path

1. Absolute path
2. Relative path

## \* Absolute path

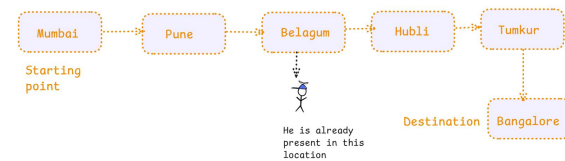


It will start from top directory (/) and provide the full location of any files or directories  
--> It will always start with a forward slash (/)

Eg: /Mumbai/Pune/Belagum/Hubli/Tumkur/Bangalore

## \* Relative path

It is relative to the current directory  
--> It will not start with a forward slash



Eg: Hubli/Tumkur/Bangalore

# File Management

\* File management is a process of storing, organizing, retrieving and managing the data in a structured way

## 1. Copy

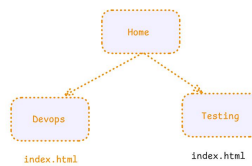
Syntax:

cp source-path destination-path

Combinations

It can be a combination of

- \* Absolute --> Absolute
- \* Absolute --> Relative
- \* Relative --> Relative
- \* Relative --> Absolute



## 2. Move

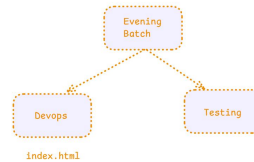
Syntax:

mv source-path destination-path

Combinations

It can be a combination of

- \* Absolute --> Absolute
- \* Absolute --> Relative
- \* Relative --> Relative
- \* Relative --> Absolute



## 3. Rename

Syntax:

mv old-filename new-filename

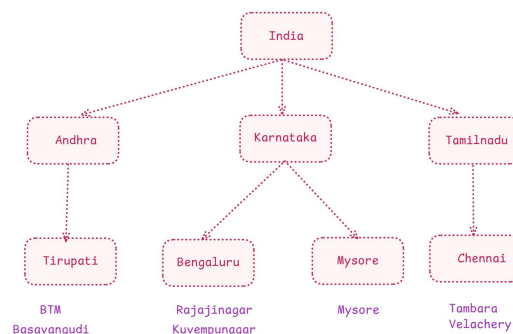
Combinations

It can be a combination of

- \* Absolute --> Absolute
- \* Absolute --> Relative
- \* Relative --> Relative
- \* Relative --> Absolute



# TASK



1. Create above directory Hierarchy
2. Move "BTM" and "Basavangudi" from "Tirupati" to "Bengaluru"
3. Copy "Basavangudi" from "Bengaluru" to "Mysore"
4. Rename "Tambara" in "Chennai" to "Tambaram"

Note: Create "Directories" and "Sub-directories" in a Single Command

# User Managemnet

- \* Managing the users in an effective way
- \* Each and everything related to users such as "creating" a new user or "removing" the user is done by this user management commands

OR

User management in Linux Operating System is a process of creating, modifying and managing the users. It will allow multiple users to access the system users

## Types of Users

1. Root/Super/Sudo  
Root user is like an administration which will have all the permissions granted for making any kind of changes in the system

2. Normal User  
Any other user which are available apart from the root user and system user is called as Normal user

Normal user is again classified into two different types

1. Default user
2. Created user

### \* Default user

- > User which has been created while setting up the machine
- > Only the default user has the permission to switch to the root user

### \* Created user

- > User which will be created after setting up the machine by default or root user is called as Created user

3. System user  
These are the predefined users created by the Operating System to run the process and the services

**Note:** The ID: 0 will be occupied the root user  
From 1-999 we will be having the system users  
Default user ID will be 1000

## Commands for user management

1. To list all the users --> `getent passwd`  
`cat /etc/passwd`
2. To check the user currently you are working on  
`whoami`
3. To check the root user  
`who`
4. To create a user  
`adduser username`
5. To switch the user  
`su username`
6. To switch the user along with its home directory  
`su - username`
7. To logout from the users  
`exit`
8. To create a user without its home/own directory  
`useradd username`
9. To delete any user  
`userdel username`
10. To delete the user along with its home directory  
`userdel -r username`
11. To change the username  
`sudo usermod -l new-username old-username`  
**Note:** -l is login name
12. To rename the user home directory  
`sudo mv /home/old-username /home/new-username`
13. To update the user Home Directory  
`sudo usermod -d /home/new-username -m new-username`  
**Note:** -d is a flag to set the home directory  
-m to move files from /home/username to username

# Group Managemnet

Whenever we want to share any of the resources within a set/group of users in such cases we can create a group

--> Group is a set of users

--> Even the group management will be done by the root/sudo/super user itself

## Types of Group

### 1. Primary Group

The group which has been created by its own user

OR

The default group a user belongs to

### 2. Secondary Group

The group which is created by other user

OR

Additional group a user is a member of

## Commands of Group management

### 1. To list all the available groups

```
cat /etc/group
```

```
getent group
```

### 2. To create a group

```
groupadd groupname
```

### 3. To add user to a group

```
usermod -aG group-name user-name
```

Note: -a --> append

-G --> Group

### 4. To delete user from the group

```
gpasswd -d username groupname
```

### 5. To delete the group

```
groupdel groupname
```

### 6. To change the group name

```
groupmod -n new-groupname old-groupname
```

### 7. To filter the specific group

```
getent group groupname
```

# File Permissions

File Permission commands helps you manage the accessibility of the files

In Linux we have 3 different set of users/entities

1. Owner/User  
Eg: For Others
2. Group  
venkata
3. Others  
index.html ← Pallavi

Permission Symbols

| Permissions   | Alphabets | Numericals |
|---------------|-----------|------------|
| read          | r         | 4          |
| write         | w         | 2          |
| execute       | x         | 1          |
| no permission | -         | -          |

Permissions that can be granted for a file

1. Read --> To see/read the content inside the file
2. Write --> To add/remove the content from the file
3. Execute --> To execute/run file

Symbols used for different entities

1. Owner/User u
2. Group g
3. Others o

To check the permissions of the file we have a command that is

--> ls -l

```
-rw-rw-r-- 1 suchith suchith 0 Jul 15 21:14 index.html
```

Diagram labels for the command output:

- rw-rw-r--: Default permissions granted to the files
- 1: hard link
- suchith: Name of the user/owner
- suchith: Name of the group
- 0: File size in bytes
- Jul 15 21:14: Last modification details
- index.html: File name

Note: The default size of the directory is "4096" bytes

-rw-rw-r-- This will be the default permissions of the file

Diagram illustrating the permission symbols for Users, Groups, and Others:

```
- r w - r w - r - -
```

Labels for the symbols:

- Users: r, w
- Groups: r, w
- Others: r

Symbol that is used to signify that it is a file: -

d will indicate the directory: d

Full permission string: d r w x r w x r - x

## Points to be remembered

1. The permissions include "read", "write" and "execute" for all the three entities
  - \* First place value (---): Users
  - \* Second (---): Group
  - \* Third (---): Others

2. In the permission set (---)

- \* 1st --> read (r)
- \* 2nd --> write (w)
- \* 3rd --> execute (x)

3. If the beginning of the permission if we are getting

- \* (-) Indicates it is a file
- \* (d) Indicates the directory

\* chmod ugo+r file\_name OR chmod a+r file\_name

This command will give the read permission for

--> user  
--> group  
--> others

"+" is used to add the permission  
"-" is used to delete the permissions

--> For giving permission for both files and directories

Syntax:

chmod permission file/directory\_name

"+" is used to add the permission  
"-" is used to delete the permissions  
"=" is used to reset the previous permissions

Eg: adding  
User --> read, write      chmod u+rw file\_name

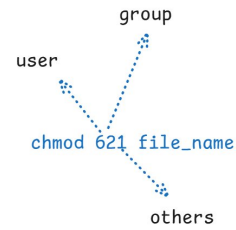
Eg: removing  
User --> read      chmod u-r file\_name

# Numericals

- \* Only read = 4
- \* Only write = 2
- \* Only Execute = 1

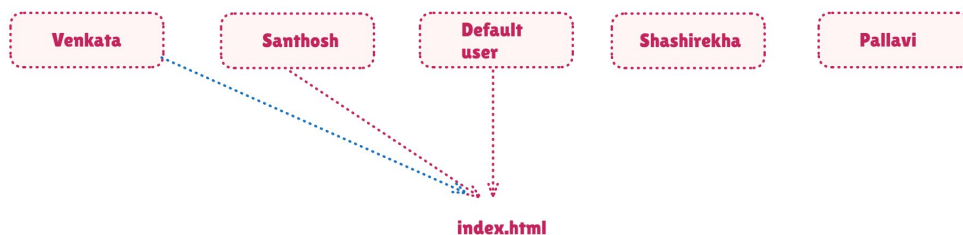
Eg: users = read + write --> 4+2 ==6 for users  
groups = write --> 2 for group  
others = execute --> 1 for others

Eg2: users = read + write + execute --> 7  
group = read + write --> 6  
others = read --> 4



| Permissions      | Users | Groups | Others |
|------------------|-------|--------|--------|
| * 771            | all   | all    | x      |
| * 000            | -     | -      | -      |
| * 124            | x     | w      | r      |
| * 447            | r     | r      | rwX    |
| * 221            | w     | w      | x      |
| * 347            | wX    | r      | rwX    |
| <del>* 815</del> |       |        |        |
| * 765            | rwX   | rw     | rx     |
| * 573            | rx    | all    | wX     |
| * 444            | r     | r      | r      |
| * 357            | wX    | rx     | rwX    |
| * 660            | rw    | rw     | -      |
| * 171            | x     | all    | x      |

# TASK



## \* chown

Change the ownership or owner of the file

--> Ownership of the file can be changes only by the root user or the actual user of the file

**chown** <new-username> file\_name

## \* chgrp

Change the group of the file

--> Group can be changed either by the owner of the file or any user which is member of that group

--> Also the root user has the permisson to change the group

**chgrp** <new-groupname> file\_name

# File Compression

File Compression reduces the size of the file by encoding their data more efficiently

Ways to compress a file

## 1. tar

- > tape archive
- > It is used to collect multiple files in a single archive file
- > It will combine multiple files and create a single file
- > Extension for the tar file is--> .tar

Commands:

--> To create a "tar" file

```
tar cvf Archive-name.tar File1 File2 File3
```

--> To extract files from a "tar" file

```
tar xvf Archive-name.tar
```

--> To extract a particular file from a "tar" file

```
tar xvf Archive-name.tar File1 File2
```

--> To list the files inside the "tar" file

```
tar -tvf Archive-name.tar
```

## 2. zip

- > It will combine multiple files and will create a single compressed file
- > Extension of zip file is .zip

Commands:

--> To create a zip file

```
zip Archive-name.zip File1 File2 File3
```

--> To extract a zip file

```
unzip Archive-name.zip
```

--> To list the files inside the "zip" file

```
unzip -l Archive-name.zip
```

--> To extract a particular file from a "zip" file

```
unzip Archive-name.zip File1 File2
```

## 3. gzip

- > It is used to compress/reduce the file size of already compressed file
- > The extension of gzip file --> .gz

--> To create a gzip file

```
gzip Archive-name
```

--> To decompress the gzip file

```
gzip -d Archive-name
```

OR

```
gunzip Archive-name
```

--> To see the files inside the gzip archive file

```
gzip -l Archive-name
```

## TASK-1

1. Create a Directory named "Project" in your home directory
2. In "Project", Create 3 subdirectories named "Docs", "Code" and "Config"
3. Create a file named "notes.txt" in "Docs" directory and add some text to it
4. List the content of "Docs" directory
5. Create a new user named "projectuser"
6. Create a new group named "projectgroup"
7. Add projectuser to the projectgroup
8. Move the "notes.txt" from "Docs" directory to the "Code" directory
9. Copy the "notes.txt" file from "Code" directory to the "Config" directory
10. List the contents of the "Code" Directory to verify the presence of notes.txt
11. Delete the "Config" directory and all its contents
12. Verify the "Config" directory and its contents are no longer present

## TASK-2

1. Create 3 users - user1, user2, user3
2. In each user directory create 2 folders--> Folder1, Folder2
3. Inside each of the folder create 2 text files each and add content in all the text files
4. Check the permissions of each file
5. Try to copy 1 file from user1 (Folder1) to user2 (Folder2)
6. Try to move the files from user3 (Folder3) to user1 (Folder2)
7. Create a group called "devops"
8. Add all the three users to the group
9. Change the group of any one file from user2 (Folder2) so that other user can access it
10. Create a user4
11. user4 should have all the permissions to read the file which are present under user2
12. user4 should have only only the write permission to read the file present under user1
13. user should have no permissions granted to any of the resources present under user3
14. change the owner of the file in user2 (Folder2) to user4

# File Systems

A file system is a method used by an operating system to store, organize retrieve and manage data on storage devices such as hard drive, SSD and USB drives

--> It provides structured way to store and access the data

## Features of file systems

### 1. File management

Stores and organize the files

### 2. File Operations

Create, Read, Write and Delete the files

### 3. Directory Management

Create and manage the folders

### 4. Space management

Allocates and track the disk space

## Types of File System

### 1. ext (Extended File system)

It has multiple versions

--> Versions

- \* ext1
- \* ext2
- \* ext3
- \* ext4

It is used to manage the storage

### 2. vfat (Virtual file allocation table)

Mainly used in Virtual drives, SD cards and USB drives

### 3. zfs (Zeta byte file system)

It is mainly used in servers and data centers

### 4. xfs (Extended File System)

It is commonly used in servers, datacenters and high speed storage system

### 5. tmpfs (Temporary file system)

It is mainly for temporary storage

--> The "Command" to check the file system in Operating System is

```
--> df : disc free          df -T
T: Type                    OR
h: Human readable format   df -Th
```

By using this command we can get multiple details of the machine such as Disk space, usage available etc

--> In one machine we can have multiple file system

# Networking Commands

--> Before using the networking commands install the net tools

```
sudo apt install net-tools
```

## Commands

### 1. ifconfig

--> Interface Configuration

--> It displays information regarding the network

\* If you down the interface

```
>> sudo ifconfig <interface-name> down
```

\* If you up the interface

```
>> sudo ifconfig <interface-name> up
```

### 2. ping hostname/ip address

--> It is used to check the reachability of specified hostname or ip address

### 3. ip address

--> It is used to display the complete details of the ip address

### 4. route

--> It shows the route tables

### 5. dig

--> It is used to fetch the information related to the host

Eg: dig flipkart.com, dig google.com

### 6. netstat

--> By using this command we can get the network statistics

### 7. wget

--> Download the resource from the given URL

### 8. curl hostname/ipaddress

--> Fetch the html content of website

### 9. man ls

--> To view the usermanual of any command

### 10. ss (Secure Socket)

--> Show all the active sockets [ TCP, UDP, UNIX ] etc

### 11. nslookup

--> It is used to fetch the data related to the specified hostname

Eg: nslookup google.com

### 12. traceroute google.com

--> To trace the packets sent or received

# Disk Utility Commands

### 1. free

--> To display the amount of free and used memory

### 2. free -h

--> Display the amount of free and used memory in the human readable format

### 3. lsblk

--> It is used to get the block devices

### 4. sudo fdisk -l

--> To display the partition information

### 5. df

--> It shows available and used space on the file system

### 6. df -ah

--> It shows available and used space of all the hidden files in the human readable format on the system

### 7. du

--> To estimate the file space usage

## Additional Commands

1. To display the hostname  
`hostname`
2. To change the hostname  
`sudo hostnamectl set-hostname <new-hostname>`  
  
OR  
  
`sudo hostname <new-hostname>`  
--> bash (It is used to refresh the terminal)
3. To get the top order of the file  
--> `head <file-name>`  
--> `head -n 5 <file-name>` (It is used to specify the line number from the top)
4. To get the bottom order of the file  
--> `tail <file-name>`  
--> `tail -n 7 <file-name>` (It is used to specify the line number from the bottom)
5. To create a shortcut of long command  
--> `alias <alias-name>=<command>`
6. Sort in ascending order  
--> `sort <file-name>`
7. Sort in descending order  
--> `sort -r <file-name>`
8. To eliminate the repeated lines  
--> `uniq <file-name>`
9. To get the count of the repeated lines  
--> `uniq -c <file-name>`  
  
--> `sort <file-name> | uniq -c`
10. Read a file and search for a keyword  
--> `less <file-name>`  
  
/<keyword>  
  
To exit from the terminal press q
11. To view the content of the file page by page  
--> `more <file-name>`  
  
Note: To move the page forward press "space"  
--> To move the page backward press "b"
12. Split the lines of the file and create a separate file  
--> `split -l <line-number> <file-name>`
13. To find the lines that match a word  
--> `grep "<keyword>" <file-name>`
14. To search multiple words  
--> `egrep "word1|word2" <file-name>`
15. Wildcard in Linux to create multiple files  
--> `touch file{1..10}`
16. To count the number of lines in a file  
--> `wc -l <file-name>`
17. To shuffle the content of the File  
--> `shuf <file-name>`
18. To check how long the server has been active  
--> `uptime`
19. To record your activity on terminal in a file  
--> `script`  
  
To save the script type "ctrl+d"  
  
Read the content of the file that is typescript
20. To calculate using Ubuntu OS  
--> `bc`
21. Find a file in linux/ubuntu (To install the tool plocate)  
--> `sudo apt install plocate`  
--> Create a file  
--> `locate <file-name>`  
  
If the file path is not displayed update the database using the command  
  
\* updatedb